ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4

Question 1

Okay, let me start with Q1 since it looks the easiest. I will go slowly and write everything out.

Given: sensor 1280 px wide, pixel pitch p = 5 μm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\text{ m})/(5\times10^{-6}\text{ m}) = 3200$. The metres cancel so the answer is in pixels. **$f_x = 3200$ px**.

(ii) Width of the sensor = N·p = 1280 × 5 μm = 6400 μm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\text{ mm})(400\text{ mm})/(16\text{ mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 μm** on the belt.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisaton sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\text{ V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Sence u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substitutting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Question 2

Hmm, this is the pendulum question. I remember doing something similar in class so lets see.

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - b\,x_2 - mg\ell\sin x_1)/J$.

At equilibrium $x_2 = 0$ and $\sin\theta_e = \tau_0/(mg\ell)$, so $0 \le \tau_0 \le mg\ell = 2.45$ N·m.

(b) The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell\sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ (s^{-2}). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.

A = [0, 1; −61.31, −2.5], and since $\partial f_2/\partial \tau = 1/J = 50$, **B = [0; 50]**.

Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stabel oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$: $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss} = 2.4525 \sin\theta_{ss}$. At $\theta_d = 90°$ the right side ≠ 0 while the left side = 0, so the joint cannot rest at θ_d.

With $\sin\theta_{ss} \approx 1$: $\theta_d - \theta_{ss} = 2.4525/50 = 0.00491$ rad = **0.281°**.

(d) Stedy state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and **$\theta_{ss} = \theta_d$: steady-state error is zero**.

Poles: linearise the closed loop about θ_d. Before compensaton, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell \cos\theta_d)\delta\theta = 0$, characteristic polynomal $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell \cos\theta_d$. After compensation the added term $mg\ell \sin\theta$ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell \cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because cos 90° = 0; if θ_d were, say, 60°, the compensaton would raise the stiffness by 61.3 × 0.02... i.e. it would move the poles.

(not 100% sure about the last part, moving on)

Question 3

This one looks long but I think I know the observer part.

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) Reason 1 – noise amplification: the backward diffrence $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequncy (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefor appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a)\, z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a)\, z^{-1}]$.

Dividing through by (1 + a): $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: **$y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\, y_{k-1}$**. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): $y = 2(0.05912)u + 0.88177\, y \Rightarrow y(1 - 0.88177) = 0.11824\, u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and C = [1, 0], $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characterstic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\, s + (2\ell_1 + \ell_2)$.

Both estimator poles at $s = -10$ means the desired polynomal is $(s + 10)^2 = s^2 + 20\, s + 100$. Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

$L = [18, 64]^T$. Verification: $A - LC = [-18, 1; -64, -2]$ has trace -20 and determinant $(-18)(-2) - (1)(-64) = 36 + 64 = 100$, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) The observer is more accurate than the filter.

The filter is safer when you need something simple and cheap to implement.

Question 4

Last question, only a few minutes left so I might rush a bit.

Given: plant $\theta^{\cdot\cdot} + a\theta^{\cdot} + c\theta = b\, u$ with $a = 2$, $c = 6$, $b = 1$; controler $u = N_{ff}\, r - k_1\theta - k_2\theta^{\cdot}$.

(a) Substitute the control law into the plant: $\theta^{\cdot\cdot} + a\theta^{\cdot} + c\theta = b\, N_{ff}\, r - b\, k_1\, \theta - b\, k_2\, \theta^{\cdot}$. Moving the feedback terms to the left: $\theta^{\cdot\cdot} + (a + b\, k_2)\, \theta^{\cdot} + (c + b\, k_1)\, \theta = b\, N_{ff}\, r$.

Numerically, $a + b\, k_2 = 2 + 2 = 4$ and $c + b\, k_1 = 6 + 2 = 8$, so the characterstic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stabel; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\, N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, **$N_{ff} = 8$**. (Note that this is larger than 1 because the graviy-related stiffness c and the feedback k_1 both resist the displacement.)

(b) $G_{DC} = bN_{ff}/(c + bk_1)$. Numerator: $0.75 \times 8 = 6$. Denominator: $c + k_1 = 8$ as before.

$\theta_{ss} = 6/8 =$ **0.75**, so error = **25%**.

(c) Substituting u and differentiating: the integral term enters with a minus sign, so the characteristic polynomial is $s^3 + (a + bk_2)s^2 + (c + bk_1)s - bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

So $2 + k_2 = 8 \Rightarrow k_2 = 6$; $6 + k_1 = 24 \Rightarrow k_1 = 18$; $-K_I = 32 \Rightarrow$ **$K_I = -32$**.

(d) The integrator keeps accumulating the error until the error becomes zero, so the steady-state error is zero whatever b and c are, if the loop is stable.

Failure mode: integrator windup when the acuator saturates. Remidy: anti-windup, e.g. clamp the integrator.

Sorry, running out of time. Hope partial marks are given for the method.